Section 3: proposed manuscript corrections

Unapplied review draft

All text below is proposed. The working annotated manuscript is unchanged. Numbering refers to the current manuscript; section 3 now occupies printed pages 31–35 after the approved section 2 expansions.

Remark 3.15 (`GLZ:char`), R050. Replace only "$\mathbb{Z}$-linear" in the first sentence by "piecewise $\mathbb{Z}$-affine". Keep the restriction on translation constants and the subsequent statement about piecewise $\mathrm{GL}_k(\mathbb{Z})$ transformations.

Independent check: on the nonzero RV group the map

$$t \longmapsto \begin{cases} t & \mathrm{vrv}(t) \geq 0, \\ t^{-1} & \mathrm{vrv}(t) < 0 \end{cases}$$

is definable and contracts to $\gamma \mapsto |\gamma|$. This is not a single affine function on a nontrivial ordered divisible group. If $|\gamma| = n\gamma + c$ everywhere, substituting 0 gives $c = 0$; substituting a and $-a$ for $a > 0$ then gives $a = na = -na$, a contradiction. This counterexample concerns functions; it need not be a bijection.

After Notation 3.21 (`RVcat:tag`), R053. Immediately before "The map $\mathbb{L}$ induces a surjective semiring homomorphism", insert:

> Suppose that $\mathbb{S}$ is (VF, Γ)-generated.

The same hypothesis already appears in Theorem 3.22. The insertion makes it apply to the preceding assertions about both lift homomorphisms. The definition of the lift on individual objects needs no such hypothesis.

The referee's particular example uses a trivially valued ground field and therefore does not meet the newly approved standing hypothesis. Here is an independent replacement. Let F be a small algebraically closed *nontrivially* valued subfield. Choose $\tau \in \mathrm{RV}$ with $\mathrm{vrv}(\tau) \notin \Gamma(F)$, and use the real parameter structure generated by F and τ. The VF part of its definable closure is F, by Lemma 2.13 and algebraic closedness of F. Hence $\tau^\sharp$ has no definable VF point. The singletons $\{\tau\}$ and $\{1\}$ are isomorphic objects of $\mathrm{RV}[1]$, whereas their lifts cannot be definably bijective: $1^\sharp$ contains the definable point 1. Definable bijections preserve the existence of a definable point, in both directions. The ground value group remains nontrivial.

Notation 3.20, R051. Correct the spellings "nonhomogenous" and "homogenous" to "nonhomogeneous" and "homogeneous".

Theorem 3.22 (`main:prop:HK`), R054. In the measured characterization, replace $\int_+$ by $\int_+^\mu$. Retain the measured congruence.

Formal verification boundary. The Lean implementation keeps the object lift separate from descent to isomorphism classes, uses full original-base function graphs, and keeps the measured relation $\mathbf{P} = 0$ separate from the unmeasured relation $\mathbf{P} - 1 = 0$. The construction of all Grothendieck semirings and the two full HK presentations is ongoing. A generic algebraic quotient lemma or clean axiom list alone does not establish those paper statements.